

# Complex Analysis – 1

1 сентября 2026 года

## Теоретический минимум.

**Опр.** For  $z = x + iy$  put  $\bar{z} = x - iy$  and  $|z|^2 = z\bar{z}$ . On the unit circle  $\bar{z} = 1/z$  — this single substitution turns most identities of this sheet into algebra. The  $n$ -th roots of unity are  $\varepsilon^k$ ,  $\varepsilon = e^{2\pi i/n}$ , and  $\sum_{k=0}^{n-1} \varepsilon^{km}$  equals  $n$  if  $n \mid m$  and 0 otherwise.

**Теорема 1 (Гаусс–Люка)** *All roots of  $p'$  lie in the convex hull of the roots of  $p$ . In particular, if all roots of  $p$  lie in a half-plane, disc or convex polygon, the same holds for the roots of  $p'$ .*

**Теорема 2 (Энестрём–Какея)** *If  $0 < a_0 \leq a_1 \leq \dots \leq a_n$ , then every root of  $a_0 + a_1z + \dots + a_nz^n$  satisfies  $|z| \leq 1$ ; if the inequalities are strict, all roots lie in  $|z| < 1$ . (Доказательство: рассмотрите  $(1 - z)p(z)$ .)*

**Теорема 3 (принцип максимума для многочленов)**  $\max_{|z| \leq 1} |p(z)| = \max_{|z|=1} |p(z)|$ , and if  $|p|$  is constant on the unit circle, then  $p(z) = cz^k$ .

**Факт 1** Виета: for  $p(z) = a_n \prod (z - z_k)$  one has  $\sum z_k = -a_{n-1}/a_n$  and  $\prod z_k = (-1)^n a_0/a_n$ ; together with  $|z| = 1 \Leftrightarrow \bar{z} = 1/z$  this settles most questions about roots on the circle.

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1. (Putnam 2005) Let  $p(z)$  be a polynomial of degree  $n$  all of whose zeros have absolute value 1. Put  $g(z) = p(z)/z^{n/2}$ . Show that all zeros of  $g'(z) = 0$  have absolute value 1.
2. (Putnam 2014) Let  $P_n(x) = 1 + 2x + 3x^2 + \dots + nx^{n-1}$ . Prove that  $P_j(x)$  and  $P_k(x)$  are relatively prime for all positive integers  $j \neq k$ .
3. (Putnam 2018) Let  $n$  be a positive integer and let  $f_n(z) = n + (n - 1)z + (n - 2)z^2 + \dots + z^{n-1}$ . Prove that  $f_n$  has no roots in the closed disc  $\{z \in \mathbb{C} : |z| \leq 1\}$ .

4. (Putnam 2019) Given real numbers  $b_0, b_1, \dots, b_{2019}$  with  $b_{2019} \neq 0$ , let  $z_1, \dots, z_{2019}$  be the roots of  $P(z) = \sum_{k=0}^{2019} b_k z^k$  and let  $\mu = \frac{|z_1| + \dots + |z_{2019}|}{2019}$ . Determine the largest constant  $M$  such that  $\mu \geq M$  for all choices of  $b_0, \dots, b_{2019}$  satisfying  $1 \leq b_0 < b_1 < \dots < b_{2019} \leq 2019$ .
5. (Putnam 2020) For  $j \in \{1, 2, 3, 4\}$  let  $z_j$  be a complex number with  $|z_j| = 1$  and  $z_j \neq 1$ . Prove that  $3 - z_1 - z_2 - z_3 - z_4 + z_1 z_2 z_3 z_4 \neq 0$ .